NIOS Class 10 (211) Mathematics — 7-Day Fast-Track Study Plan

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

NIOS Class 10 (211) Mathematics — 7-Day Fast-Track Study Plan

Designed for Working Students | Exam Date: 5th November

> **Target**: Pass comfortably with **50+ to 65+ marks** out of 85.
> **Daily Time Required**: 2 hours on weekdays (1 hr morning + 1 hr night), 3–4 hours on weekend.

The Working Student's Realistic Routine

- **Morning Sprint (45–60 mins)**: Review formulas, learn 1 high-yield template, solve 5 MCQs.
- **Commute / Breaks (15–20 mins)**: Glance through the `02_ALL_IN_ONE_FORMULA_CHEAT_SHEET.md` on your phone.
- **Evening Session (60–90 mins)**: Practice the target chapter's solved questions with pen and paper.

Day-by-Day Roadmap (Oct 30 – Nov 5)

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    D1["Day 1 (Oct 30): Guaranteed 17 Marks<br>Heights & Distances + Trig Values + Probabilit
    D2 --> D3["Day 3 (Nov 1): Geometry & Graph (15 Marks)<br>Graphical Equations (5M) + Circl
    D3 --> D4["Day 4 (Nov 2): Statistics & Coordinate Geom (15 Marks)<br>Mean/Median Formula
    D4 --> D5["Day 5 (Nov 3): Mensuration & Commercial (12 Marks)<br>Cylinder/Cone/Sphere + I
    D5 --> D6["Day 6 (Nov 4): Objective Blitz & Mock Test<br>Practice Top 50 Objectives (42M)
    D6 --> D7["Day 7 (Nov 5): Exam Day Final Warm-Up<br>Formula Cheat Sheet Review & Confiden
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Day 1 (Friday, Oct 30): The Easiest 17 Marks

- **Topics**:
- **Trigonometric Values Table** ($\sin, \cos, \tan$ for $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$).
- **Heights & Distances (5 Marks Sure-Shot!)**: Tower, light house, angle of elevation/depression.
- **Probability (Ch 26 - 5 Marks)**: Coins, 1 or 2 dice, cards, balls in a bag.
- **Goal**: Complete these 2 topics tonight. You will have locked in **17 marks** on Day 1!
- **Read**:
- `04_GUARANTEED_SURE_SHOT_LONG_QUESTIONS.md` (Template 1: Heights & Distances)
- `02_ALL_IN_ONE_FORMULA_CHEAT_SHEET.md` (Trigonometry & Probability section)

Day 2 (Saturday, Oct 31): Algebra Powerhouse (15 Marks)

- **Topics**:
- **Arithmetic Progression (AP - Ch 7)**:
  - n-th term: $a_n = a + (n-1)d$
  - Sum: $S_n = \frac{n}{2}[2a + (n-1)d]$
- **Quadratic Equations (Ch 6)**:
  - Discriminant $D = b^2 - 4ac$ (Nature of roots: $D > 0, D = 0, D < 0$)
  - Finding roots using formula $x = \frac{-b \pm \sqrt{D}}{2a}$
- **Special Products & Factorisation (Ch 4)**:
  - $(a+b)^2, (a-b)^2, a^2 - b^2, a^3 + b^3, a^3 - b^3$.
- **Goal**: Solve the 8 standard problems in `05_CHAPTER_BY_CHAPTER_SUPER_SUMMARY.md`.

Day 3 (Sunday, Nov 1 - Weekend Deep Work): The 5-Mark Graph & Geometry (15 Marks)

- **Topics**:
- **Linear Equations Graphical Solution (5 Marks Sure-Shot!)**:
  - Plotting $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ on a graph sheet, finding intersection point (x, y).
- **Circle Tangents & Angles (Ch 16 & 17)**:
  - Tangent $\perp$ radius ($OP \perp PT$).
  - Tangents from external point are equal ($PA = PB$).
  - Angle in semicircle is 90°; Cyclic quadrilateral opposite angles sum to 180°.
- **Construction (Ch 18 - 3 to 5 Marks)**:
  - Tangents to a circle from an external point using compass and ruler.
- **Goal**: Practice drawing 2 graphs and 2 circle tangent constructions on paper!

Day 4 (Monday, Nov 2): Statistics & Coordinate Geometry (15 Marks)

- **Topics**:
- **Statistics (Ch 25)**:
  - Grouped data Median: $L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$
  - Grouped data Mean: $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$
  - Empirical relation: $\text{Mode} = 3\text{Median} - 2\text{Mean}$
- **Coordinate Geometry (Ch 19)**:
  - Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
  - Mid-point formula: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
  - Section formula: $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$
- **Goal**: Solve 2 median problems and 3 coordinate geometry problems.

Day 5 (Tuesday, Nov 3): Mensuration & Commercial Maths (12 Marks)

- **Topics**:
- **Surface Area & Volume (Ch 21)**:
  - Cylinder: $V = \pi r^2 h$, $\text{CSA} = 2\pi rh$, $\text{TSA} = 2\pi r(r + h)$
  - Cone: $V = \frac{1}{3}\pi r^2 h$, $l = \sqrt{r^2 + h^2}$, $\text{CSA} = \pi rl$
  - Sphere & Hemisphere: $V = \frac{4}{3}\pi r^3$, $\text{TSA} = 4\pi r^2$
- **Installment Buying (Ch 9)**:
  - Cash price, Down payment, Balance owed, Interest calculation.
- **Goal**: Master formula substitutions for direct 2-mark and 3-mark questions.

Day 6 (Wednesday, Nov 4): The Objective Blitz & Rehearsal (Crucial Day!)

- **Topics**:
- Go through `03_TOP_50_REPEATED_OBJECTIVES_MCQ_SOLVED.md` from start to finish.
- Attempt the 85-mark `06_PREDICTED_MOCK_TEST_PAPER_2026.md` under timed conditions (or self-check answers).
- Pack your exam kit: Admit card/hall ticket, ID proof, Geometry box (ruler, compass, sharp pencil, eraser), black/blue pens.

Day 7 (Thursday, Nov 5 - Exam Day Morning):

- **45-minute calm morning revision**:
- Read through `02_ALL_IN_ONE_FORMULA_CHEAT_SHEET.md`.
- Remember: You only need 28 to pass, but your target is 50–65+.
- In the exam room: First 15 minutes, tick all the objective questions you know directly, then solve the 5-mark Heights & Distance and Graphical Linear equation. Walk in with full confidence!